

An Inexpensive 5-COLOUR LED FLASHER

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This 5-colour LED flasher circuit can be built using the popular 555 timer and 4017 decade counter ICs. It does not require any expensive component like a microcontroller.

The timer IC is used as an astable multivibrator to generate a clock pulse, which is fed to the clock input of the 4017. Outputs of the 4017 are connected to bases of five transistors to drive 20 LEDs of different colours, each with its own current-limiting resistor.

The 4017 advances to the next

output every time it receives a clock pulse, causing the LEDs to turn on and off in sequence. The speed of the flashing can be adjusted by changing the values of the resistors and capacitors in the 555 timer circuit. The author's prototype is shown in Fig. 1.

Circuit diagram of the 5-colour flasher is shown in Fig. 2. The circuit comprises a 12V battery (or 12V adaptor), timer IC 555 (IC1), decade counter IC 4017 (IC2), five BC548 transistors (T1 through T5), five sets of 5mm red, green, blue, white, and

yellow LEDs each, and a few other components.

If you have built 555 timer related projects, you would be familiar with the IC. The timer IC's pin 4 (reset) and pin 8 (Vcc) are connected to the voltage supply, which can be anywhere between 9V and 12V.

In this circuit, a 10k resistor (R1) is connected between Vcc and discharge pin 7 of IC1. A 1k resistor (R2) is connected between pin 7 and the junction of pin 2 (trigger) and pin 6 (threshold).

A 10 μ F capacitor is connected between pin 6 and the ground pin of 555 timer, whose ground pin 1 (GND) is connected to the circuit's ground. A 100nF capacitor is connected between pin 5 (CV) of the timer and the ground. The IC's out pin 3 is connected to clock pin 14 of IC 4017 (IC2).

Pin 16 of decade counter IC2 is connected to supply. Its pins 8 (VSS), 13 (CKEN), and pin 15 (Reset) are connected to ground. The

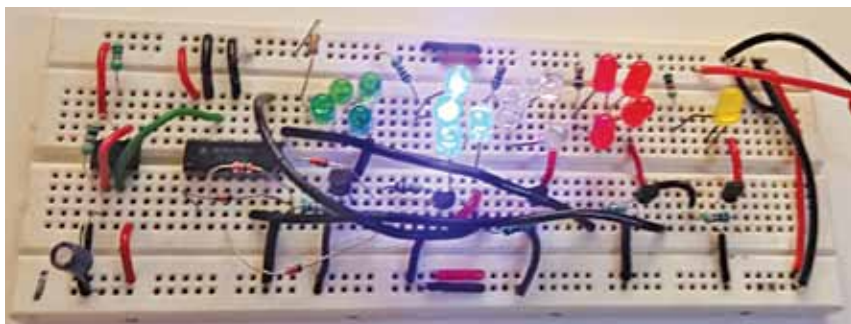


Fig. 1: Author's prototype on a breadboard

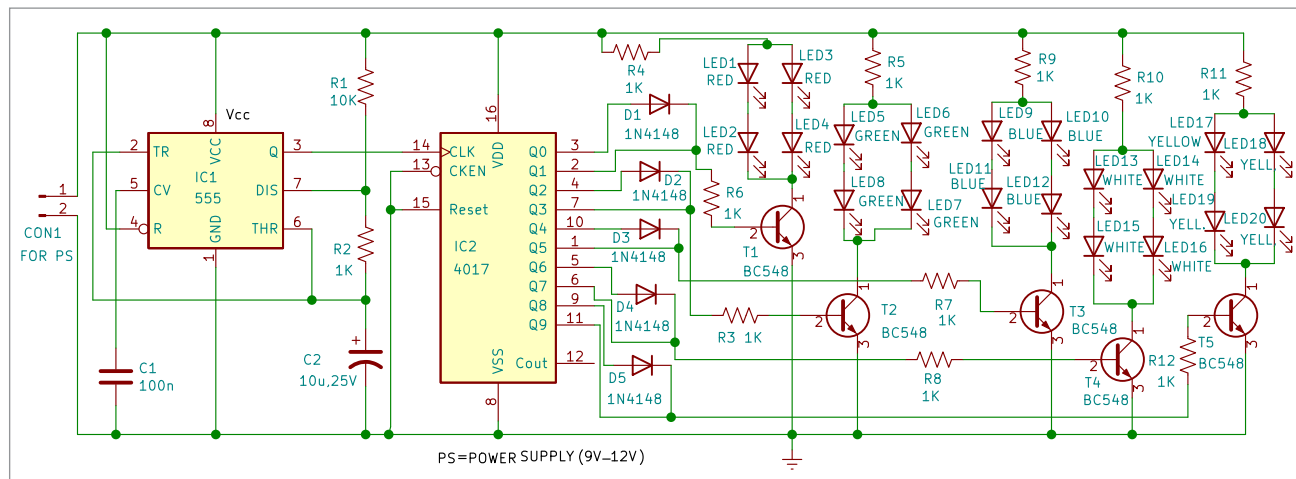


Fig. 2: Circuit diagram